# 5

# *Infrastructure-as-a-Service (IaaS) Delivery Model in Cloud*

## Learning Objectives

After reading this chapter, you will be able to

- Describe the need for Infrastructure-as-a-Service (IaaS)
- Explain the applications of IaaS
- Compare and contrast IaaS services from several Cloud service providers
- Identify Challenges of IaaS

Massive data centers and server farms comprise the Cloud's underlying infrastructure on which Software-as-a-Service (SaaS) and Platform-as-a-Service (PaaS) run. It is an emerging computing paradigm where data and services reside in massively scalable data centers and can be ubiquitously accessed from any connected devices over the Internet. The capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.

**Preliminaries** The following are some of the terms and terminologies defined here for easier understanding in the remaining part of this chapter.

**Virtual Machine** (VM): The VM is the basic unit of computing in the system. VMs come in two flavours: non-persistent and persistent. A non-persistent machine does not persist anything after the VM is stopped.

Everything on the machine that is modified while it is running is lost. In contrast, a persistent machine is backed by permanent storage (likely a virtual disk) that continues to exist after the machine is stopped. This allows the machine to be restarted into the same state that it was in the last time it was stopped.

**Virtual Disk**: A virtual disk is size-configurable, permanent, block-level storage that can be mounted to a running VM. Virtual disks are capable of random I/O. Virtual disks can only be mounted to a single VM at any time, but can be mounted to multiple machines throughout its lifetime.

**Geographic Region:** Despite the term "virtual", the machines and disks do physically exist somewhere. The geographic region is the place where the resources powering these virtual entities physically reside.

**Failure-insulated Zone:** These are sub-divisions of the geographic regions. Geographic regions are useful when thinking about large-scale failures and disasters (earthquakes, explosions, hurricanes) that cover a large area. Failure-insulated zones are divisions within a geographic region that are, as much as possible, isolated from expected localized failures such as disk or power supply failure.

**Archival Storage:** It is long term, permanent, blob-level storage. It allows the storage and retrieval of individual blobs, but does not allow random I/O within the blobs. Archival storage is not mounted to any VM, and can be accessed by multiple VMs at the same time. It exists outside of any specific geographic region. It is considered to be completely durable, but not always available.

**Figure 5.1** shows IaaS, PaaS and SaaS layer. IaaS businesses offer services such as pay-as-you-go storage, networking, virtualization etc. A PaaS vendor provides hardware and software tools over the Internet, and people use these tools to develop applications. SaaS platforms make software available to users over the Internet, usually for a monthly subscription fee (e.g, email application, Customer Relationship Management (CRM) etc.). **Figure 5.2** shows Cloud IaaS components, such as hardware, data centers, bandwidth, load balancers, virtual server space, and Cloud hosting. The consumers of Cloud services looks in for either of these compo-

nents for running their applicaations in virtualized environment of the Cloud. This chapter looks at the general constituents of IaaS. The management of virtual machines and IaaS service providers are also covered.

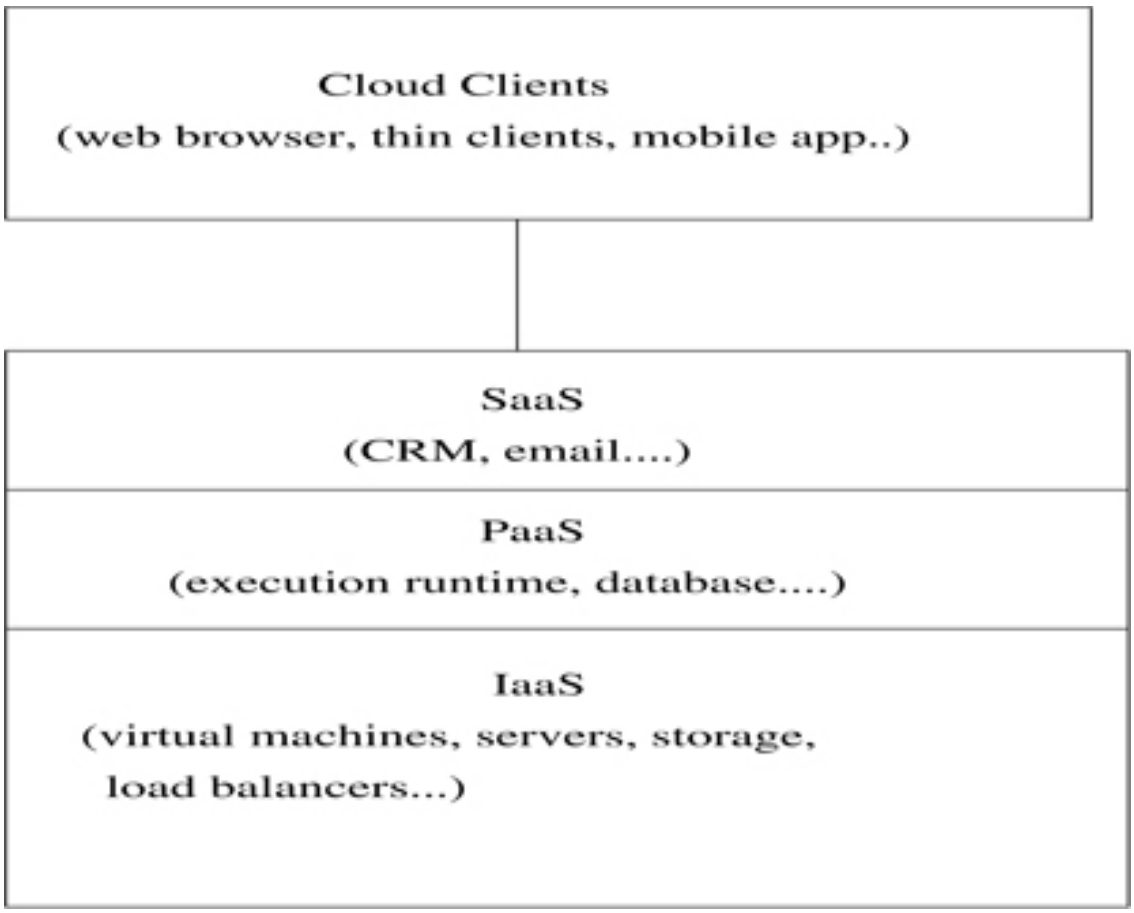


FIGURE 5.1
SaaS, PaaS, and IaaS

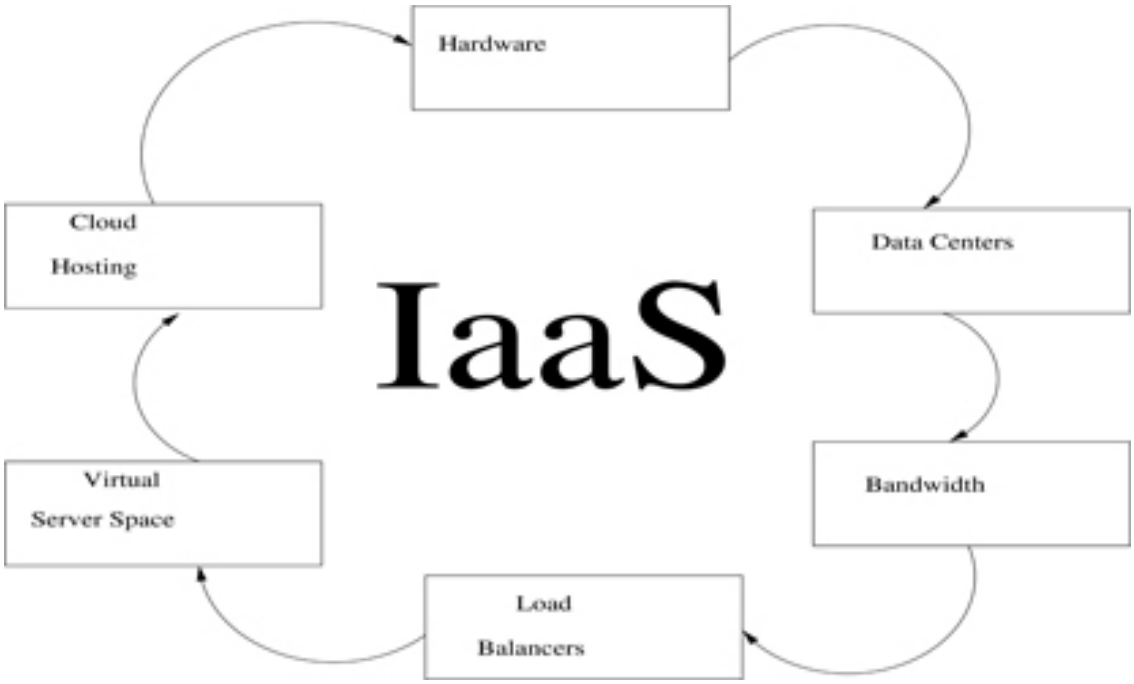


FIGURE 5.2
Components of an IaaS

## 5.1 IaaS in Cloud

To help understand the IaaS delivery model, the following sections examine some of its key characteristics, including dynamic scaling, agreed-upon service levels, renting, licensing, metering and self-service. All of these characteristics are the same in both public and private IaaS Cloud environments.

**Dynamic scaling:** Some level of uncertainty always exists when planning for IT resources. One of the major benefits of IaaS for companies faced with this type of uncertainty is the fact that resources can be automatically scaled up or down based on the requirements of the application. This important characteristic of IaaS is called dynamic scaling. If customers wind up needing more resources than expected, they can get them immediately (probably up to a given limit). A provider or creator of IaaS typically optimizes the environment so that the hardware, the operating system and automation can support a huge number of workloads.

**Service levels:** Consumers acquire IaaS services in different ways. Many consumers rent capacity based on an on-demand model with no contract. In other situations, the consumer signs a contract for a specific amount of storage or compute. A typical IaaS contract has some level of service guarantee. At the low end, a provider might simply state that the company will do its best to provide good service. If the consumers are willing to pay a premium price, they might get a mirrored service so that there are almost no change-of-service interruptions. A typical service-level agreement states what the provider has agreed to deliver in terms of availability and response to demand. The service level might, for example, specify that the resources shall be available 99.999% of the time and that more resources shall be provided dynamically if greater than 80% of any given resource is being used.

**The rental model:** When companies use IaaS, it is often said that the servers, storage, or other IT infrastructure components are rented for a fee based on the quantity of resources used and how long they are in use. Although this is true, there are some important differences between this

rental arrangement and the traditional rental models that we are familiar with. For example, when we purchase server and storage resources using IaaS services, we gain immediate virtual access to the resources we need.

Within a private IaaS model, renting takes on a different focus. Although Cloud service providers (CSPs) might not charge each user to access a resource, in the charge-back model, they can allocate usage fees to an individual department based on usage over a week, month, or year. Because of the flexibility of the IaaS model, they can charge more of the budget to heavy users.

**Licensing:** The use of public IaaS has led to innovation in licensing and payment models for software we want to run in these Cloud environments. For example, some IaaS and software providers have created a "bring your own license" (BYOL) plan so that we have a way to use our software license in both traditional and Cloud environments.

Another option is called "pay as you go" (PAYG), which generally integrates the software licenses with the on-demand infrastructure services. For example, say that you are running Microsoft Windows Server and using the PAYG route. If you are paying ten cents an hour for Cloud access, a few cents of that fee might be going to Microsoft.

**Metering and costs:** We derive a potential economic benefit by controlling the amount of resources we demand and pay for it, so that we get the right match for our requirements. To ensure that users are charged for the resources they request and use, IaaS providers need a consistent and predictable way to measure usage. This process is called metering. Ideally, the IaaS provider will have a transparent process for identifying charges incurred by the user.

With multiple users accessing resources from the same environment, the IaaS provider needs an accurate method for measuring the physical use of resources to make sure each customer is charged the right amount. In addition to the basic per-instance charge discussed below, Amazon EC2 IaaS provider includes charges such as the following (note that charges can fluctuate depending on market scenarios):

1. Storage: A per-gigabyte (GB) charge for the persistent storage of data of approximately $0.00015 an hour or $0.10 a month.
2. Data transfer: A per-GB charge for data transfers of approximately $0.15. This fee may drop to $0.05 per GB, if we move large amounts of data during the billing month. Some providers offer free inbound data transfers or free transfers between separate instances on the same provider.
3. Optional services: Charges for services such as reserved IP addresses, virtual private network (VPN), advanced monitoring capabilities, or support services. Because of the variety of costs, many Cloud providers offer cost calculators, so that organizations can estimate their monthly IaaS costs. The typical billing options are as follows:
    1. Pay as you go: Users are billed for resources used based on instance pricing.
    2. Reserved pricing: Users are billed an upfront fee, depending on the instance type, and then billed at a discounted hourly usage rate. For some companies, this provides the best rates.
    3. Trial use: Some IaaS providers, such as Amazon, offer free usage tiers that allow users to “try before they buy”.

## 5.2 IaaS Benefits

IaaS is a good solution to the problem in many ways. The top five reasons for using IaaS are: cost savings on hardware and infrastructure, capacity management, disaster recovery/business continuity, cost savings on IT staffing and administration; and the ability to access new skills and capabilities. IaaS is frequently used for the following scenarios.

1. Organisation infrastructure: IaaS can include private and public networks. Private networks are only available to your organisation, while public networks are shared with the world. This allows organisations to utilise pooled networking and server resources, which they can use to run applications and store data for day-to-day business.
2. Cloud hosting: It gives a pool of virtualised workers. These can be utilized for sites including Internet-based applications, which exist on

various servers dissipated over a huge topographically scattered network.

IaaS allows organisations to create scalable, cost-effective IT solutions without the complexity and costly hardware required to do so in-house. IaaS provides lower upfront cost, predictable cost, simple scalability, increased reliability and improved physical security.

## 5.3 Management of VMs in IaaS

There are lots of virtualization management tools, but which ones are right for your infrastructure needs to be correctly decided. The steps involved in managing VM are discussed here.

### Creating and managing VMs

The servers and VMs tab is used to create and manage VMs. VMS can be created using:

- ISO files in a repository (hardware virtualized only)
- Mounted ISO files on a Network File System (NFS), HTTP or FTP server (paravirtualized only)
- VM templates (by cloning a template)
- Existing VM (by cloning it)
- VM assemblies

VM require most installation resources to be located in the storage repository with the exception of mounted ISO files for paravirtualized guests. Before you create a VM that requires network connectivity, or a paravirtualized machine which requires network connectivity to perform the operating system install, virtual network interfaces using the virtualized Network Interface Card (VNIC) Manager must be generated.

The following list provides an outline of actions that can be performed in the VM tab:

**View VM information and events:** You can view information and access VM events by viewing information and events for VMs.

**Edit a VM:** You can edit VM parameters and you can also convert a VM to use paravirtualization. This involves making changes to the VM itself. Also start, stop, kill, restart, suspend and resuming a VM process is possible.

**Move a VM between repositories:** It is possible to create alternate repositories using external storage. If you have chosen to create an alternate repository, this function can be used to move a VM from one repository to another. It is important to understand that VM hosted on alternate repositories are outside of the scope of your support agreement. Moving a VM from one server to another, to or from the Unassigned VM folder can be the next process.

**Migrate a VM:** Migration of VMs can only be achieved between servers and between a server and the Unassigned VM folder. All compute nodes use the same CPU architecture, so you do not need to modify Server Processor Compatibility Groups.

## 5.4 IaaS Providers

IaaS providers maintain the equipment, while users rent or buy the specific services that they need. The term for rental or purchase is usually “pay as you go” which offers a lot of flexibility and ease in growth. Organizations use the IaaS Cloud as a way of easily scaling new or existing applications to larger audiences. Some of the commonly known service providers are discussed here.

### 5.4.1 Amazon AWS

Amazon Elastic Compute Cloud (Amazon EC2) is an Amazon Web Service (AWS) you can use to access servers, software, and storage resources across the Internet in a self-service manner. It provides scalable, pay as-you-go compute capacity. Also the resources provided are elastic and

scales in both direction. Chief characteristics of the services provided are: (i) Amazon Machine Image (AMI) and Instance, (ii) Region & Zones, (iii) Storage, (iv) Networking and Security, (v) Monitoring, (vi) Auto Scaling and (vii) Load Balancer.

## 5.4.2 Windows Azure

Windows Azure started as a platform as a service (PaaS) provider but now offers Infrastructure-as-a-Service (IaaS) as well. In spite of its name, this is not a Windows-only IaaS; you can just as easily run Linux VM as would Windows Server instances. The compute and storage services in Windows Azure are typical of what you will find in most IaaS providers but with the ease that you can create a SQL Server database reminds us we are on a Microsoft platform. You also have ready access to virtual networks, service buses and non-relational storage platforms as well.

### Getting Started with Windows Azure

Microsoft has one of the better IaaS management interfaces with a high-level view of all your active resources, wizards to walk you through multi-step processes, and a log display that is there when you need it but does not get in your way most of the time. Chief characteristics of Windows Azure are scalability, data protection, running applications from anywhere, and smart decision-making.

## 5.4.3 Google Compute Engine

Google Compute Engine lets you create and run VMs on Google infrastructure. Compute Engine offers scale, performance, and value that allows you to easily launch large compute clusters on Google's infrastructure. There are no upfront investments and you can run thousands of virtual CPUs on a system that has been designed to be fast, and to offer strong consistency of performance.

**Google Compute Engine Pricing:** Google Compute Engine charges for usage based on the following price sheet. A bill is sent out at the end of each billing cycle, listing previous usage and charges. Prices on this page are listed in US dollars (USD). Disk size, machine type memory, and network

usage are calculated in gigabytes (GB), where 1 GB is 1073741824 bytes. This unit of measurement is also known as a gibibyte (GiB). When charging in local currency, Google will convert the prices listed into applicable local currency pursuant to the conversion rates published by leading financial institutions.

**Machine type pricing:** Google Compute Engine offers two categories of machine types: predefined machine types and custom machine types. Predefined machine types have preset virtualized hardware properties and a set price, while custom machine types are priced according to the number of vCPUs and memory that the VM instance uses.

**Machine type billing model:** The following billing model applies to all machine types, predefined or custom. All machine types are charged a minimum of 10 minutes. For example, if you run your VM for 2 minutes, you will be billed for 10 minutes of usage. After 10 minutes, instances are charged in 1 minute increments, rounded up to the nearest minute. An instance that lives for 11.25 minutes will be charged for 12 minutes of usage.

If you run an instance for a significant portion of the billing month, you can qualify for a sustained use discount. When you use an instance for more than 25% of a month, Compute Engine automatically gives you a discount for every incremental minute you use for that instance. The discount increases with usage and you can get up to a 30% net discount for instances that run the entire month. Sustained use discounts are calculated and applied to your bill as your project earns them.

**Sustained use discounts for predefined machine types:** When Compute Engine calculates use for predefined machine types, equivalently provisioned machines that run non-concurrently are treated as inferred instances. This gives you the flexibility to start and stop instances freely and still receive the maximum sustained use discount that is available across all of your instances. For example, if you run an n1-standard-1 instance for 75% of the month, your charges are calculated as, (i) first 25% is charged at the full on-demand rate, (ii) next 25% is charged at a 20% discount off the on-demand rate and (iii) last 25% is charged at a 40% discount off the on-demand rate.

**Inferred instances:** When computing sustained use discounts for predefined machine types, Compute Engine gives customer's the maximum available discount using inferred instances. An inferred instance combines multiple, non-overlapping instances in the same zone into a single instance for billing. With inferred usage, users are more likely to qualify for sustained use discounts. Non-overlapping instances running the same predefined machine type in the same zone are combined to create one or more inferred instances.

### 5.4.4 Rackspace Open Cloud

Designing a custom architecture for customer's mission-critical applications and workloads might seem complex, but it has solution specialists who can help customers every step of the way. From server, network and storage configuration, monitoring and support, to extending to the Cloud of your choice, they help making sure customers have everything their business needs. The customers have the service request with regard to: (i) Choose a base configuration from portfolio of dedicated server bundles to start tuning the hardware to meet their workload requirements, (ii) Chat with one of company's server specialists or engineers to customize the hardware for specific solutions, from virtualization to web applications, (iii) Rackspace deploys customer's server(s) in their data centers and (iv) Customize the application resources and the environment by leveraging Rackspace technical experts.

### 5.4.5 HP Enterprise Converged Infrastructure

The customers can gain more flexibility, more visibility and more freedom with the latest release of HPE Helion CloudSystem, featuring proven innovations in Cloud infrastructure, application platforms, management and hardware. The customer's business will benefit from: (i) enhanced networking, storage, servers, lifecycle management, security features, (ii) powerful tools to build and deploy apps across multiple Clouds, (iii) faster, easier, and more flexible IT infrastructure management and (iv) updated marketplace and service design.

## 5.5 Keys to Successfully Moving to IaaS

Many organisations have successfully implemented IaaS. While Amazon is currently the top IaaS provider, other companies such as IBM, Microsoft and more specialised providers like Rackspace and RightScale all have their place in the field. When it comes to moving to IaaS environment, there are five main things to consider, such as, assessment of current workload and processes, selection of a service that provides data sovereignty, review of SLAs, adequate attention with regard to storage capabilities, network bandwidth and compute power, and ensuring financial stability.

## 5.6 Challenges of IaaS

Before signing an IaaS contract, we should check the IaaS provider's information. The classic use case for Governance in Cloud Computing is when an organization wants to prevent rogue employees from misusing a service. For example, the organization may want to ensure that a user working in sales can only access specific leads and does not have access to other restricted areas. Another example is that an organization may wish to control how many virtual machines can be spun up by employees, and, indeed, that those same machines are spun down later when they are no longer needed.

We must also evaluate our regulatory compliance requirements. Usage of Cloud Services is on a paid-for basis, which means that the finance department will want to keep a record of how the service is being used. The Cloud Service Providers themselves provide this information, but in the case of a dispute, it is important to have an independent audit trail. Audit trails provide valuable information about how an organization's employees are interacting with specific Cloud services, legitimately or otherwise. The end-user organization could consider a Cloud Service Broker (CSB) solution as a means to create an independent audit trail of its Cloud service consumption.

The data center's location can play a big role in service quality and operation. A data center close to an area that frequently gets natural disaster damage may have down time. It is a good idea to research past clientele to check and see if the provider has the certifications to back up its claims. Vendor lock-in is perceived as a significant challenge to the adoption of Cloud computing. Vendors can currently mitigate this concern by making their services more transparent so that customers can understand how their resources are being managed.

## Summary

In this chapter, we focused on IaaS, that is the rental of shared and virtualised computing processing, memory, and data storage for the purpose of running an operating system such as Microsoft Windows Server or Linux. This is in contrast to the hosted software application side of Cloud computing, generally known as SaaS and PaaS, discussed in the next chapters.

## Keywords

| | |
|---|---|
| IaaS | Scalability |
| Resiliency | Virtualization |

## Objective type questions

1. What second programming language did Google add for App Engine development?

(a) C++

(b) Flash

(c) Java

(d) Visual Basic